SHORTS

### AN IMPLANTABLE ANTENNA

A prototype silk biosensor could someday alert doctors to signs of disease. Plus, it's biodegradable, pliable, tougher than kevlar, and can be implanted in the body. Will this be the secret to the next invisibility cloak?
http://bit.ly/agKajr

### TERRY CHILDS GETS FOUR YEAR SENTENCE

Former systems admin for the city of San Francisco, was sentenced to four years in prison for refusing to hand administrative passwords back to the city.
http://bit.ly/9fJPzd

### NO BLACKBERRY BAN IN SAUDI ARABIA

This one was a little obvious. Of course there would be no ban. And to all those rumour-mongers going about telling everyone that BlackBerry is going to be kicked out of India, get real, like RIM would lose a lucrative market like India
http://bit.ly/dwzowz

### INDIA TESTING WAYS TO ACCESS BLACKBERRY

Like we said. RIM is not about to lose the right to sell its devices and services in one of the the world's fastest growing telecom markets. The trick, apparently, will be to monitor decrypted mails at a BlackBerry Enterprise server
http://bit.ly/9j3Mcb

# Build a Linux Media Center PC

PCMAG DIGITAL NETWORK



Media Center PCs have been around for almost a decade—even longer if you count earlier forays like Gateway's Destination PC lineup from the mid-1990s. It's not just about PCs, either. More recently, we've seen set-top box media extenders and media-focused game consoles like the Xbox 360 and PS3, as well as targeted peripherals like the Slingbox PRO-HD and the Hauppauge HD PVR. All have garnered their share of fans.

Unfortunately, many media center setups are inflexible. The Apple TV is probably the worst example; it's tough to do much with that box unless you're a slave to the iTunes Store. (Or unless you hack it.) But even Windows 7 and the Xbox 360 can be unnecessarily limiting when it comes to your media.

That's where Linux comes in. Granted, a lot of the fun is thanks to the hobbyist nature of the OS, at least on the consumer end. There's also a distinct feeling of getting something for nothing—and in many cases, better performance than the paid options from Microsoft or Apple.

On the other hand, though Linux-based projects are enjoyable, they can take up a lot of time, and they're not always that useful once finished. Think about it: In today's market, $500 can buy a pretty nice Windows 7 desktop machine with an HDMI port and a wireless card. It wouldn't be much of a gaming box. But you'd have a pretty slick setup for your HDTV—one that's capable of storing all your media files, and one that's much easier to troubleshoot when something goes south.

Even so, choosing Linux offers a number of advantages. In this article, we're going to show you how to build a Linux-based media center that acts as a music, video, and photo server for your entire house. It will display video on a large-screen TV without a hitch. Most importantly, coupled with MythTV, the system will record TV shows and act as a genuine DVR. This last point is key, because many so-called "media center" setups don't actually do this, and just exist to play standalone media files or stream video from the Web.

In fact, Linux has come far enough—and is powerful enough—that we bet you'll want to keep using this machine even after you're done with the project.

Sound good? Let's get started.

## Choosing the Hardware

It's been five years since we last built a Linux media center PC running MythTV. Since then, we've seen DRM-encoded music virtually disappear. We've also seen the rise of high-definition cable, the widespread adoption of Linux-powered Android smartphones, and a gradual but marked improvement in usability among various popular Linux distributions.


Moreover, it has become a lot easier to set up MythTV across various distributions. The one we use in this article (Mythbuntu) already comes with almost everything you need. There's also plenty of documentation out there—probably too much. Many wiki pages point to each other. Worse, while there's tons of information, there aren't a whole lot of "do this, and it will work"-type instructions. The forums are filled with some extremely helpful Linux fans, but it's not always obvious what to do if you want to start from scratch. What follows here is one way that will work. We want to present, from start to finish, a method the first-timer can use to build a Linux media center PC.

For this build, we aimed somewhere in the middle in terms of CPU power. Linux is as efficient as ever in terms of maximizing cheaper hardware, so that didn't pose a problem at all. But we're also talking about video recording; consequently, we wanted to give this machine enough oomph that it didn't buckle while recording the season finale of House. That meant that this is one time we would steer clear of using our dusty old P4.

We confirmed that during our initial search for a TV tuner card. "This is a pretty common misconception," Hauppauge senior software engineer Devin Heitmueller told us. "People very rarely build MythTV boxes in order to save money. In fact, it is usually suggested to use modern hardware as opposed to trying to make it work on an old PC you had in the back of the closet, as you are more likely to have success."

We culled the AMD processor and Gigabyte motherboard straight from the least expensive machine in our recent "Build the Best Gaming PC for Any Budget" article. These proved just fine, with one caveat: There's no HDMI port on that motherboard.